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STUDY AND SIMULATION OF VARIOUS HEAT TRANSFER MODELS OF SI ENGINE

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ABSTRACT
Heat transfer from cylinder walls is a major factor for engine performance, emission and the efficient design of cooling system of internal combustion engines. Furthermore, heat transfer phenomenon causes thermally induced mechanical stresses compromising the reliability of engine components. The ability to predict heat transfer in engines plays an important role in engine development and it's cooling. Today, predictions are increasingly being done with numerical simulations. These methods are based on the principles of heat transfer. This paper shows a comparative study of different heat transfer models of SI engine. MATLAB programming was used here to simulate the various performance parameters of an SI engine to find the best possible model. To understand which model works better in which region of engine cycle various comparisons have been depicted graphically with available experimental results. Furthermore, some justifications of the simulations were made by showing results for different set of engine parameters.

Keywords: Engine; Heat transfer; Modeling; Simulation;

1. Introduction

Most of the research devoted to engines focuses on the combustion process. The combustion is indeed the driving event for the whole system and deep understanding of this process is the most likely way to achieve key improvements. Engine cycle modelling and simulation along with experimental validation can provide parametric and comparative performance analysis for different performance parameters at significantly lower expense of time and money. The purpose of this thesis work is to summarize notions and ideas relevant to the modeling of heat transfer in spark ignition internal combustion reciprocating engines. More specifically, it concerns the heat transferred from the thermodynamic cycle to the solid boundaries of the engine. The main intention is for it to serve as background information necessary for modeling SI engine, particularly when using two zone modeling.

2. Modelling, Simulation and Analysis

2.1 Governing Equations of the SI Engine Cycle

For SI engines the combustion chamber can be considered as a closed system during compression, combustion and expansion because these processes occur after the inlet valve closing(IVC) and before the exhaust valve opening(EVO). The basic thermodynamic equation for this case

$$\delta Q_{fuel} - \delta Q_{loss} = \delta Q_{net} = dU + \delta W \quad (1)$$

The combustion model used is based on the model developed by Ferguson [1] for the arbitrary heat release of fuel inducted engines. The control volume considered for combustion is modeled by following equation

$$m\frac{du}{d\theta} + u\frac{dm}{d\theta} = \frac{dQ}{d\theta} - P\frac{dV}{d\theta} - \frac{\dot{m}_l h_l}{\omega} \quad (2)$$

The energy within the system is split into two sets of constituents, burned and unburned. Both are assumed to have the same pressure but temperature is not necessarily the same. The burn rate is modeled using the Wiebe function given in the following equation. Constants b=5 and n=3 were used like in [2]

$$x = 1 - e^{-b\left[\left(\frac{\theta-\theta_s}{\theta_b}\right)^n\right]} \quad (3)$$

In order to determine the mass lost during combustion across the piston rings a general equation is developed. Mass loss is only considered to be a function of engine speed. To correlate the mass loss with experimental findings the constant C is introduced.

$$m = m_1 e^{\frac{C(\theta-\theta_1)}{\omega}} \quad (4)$$

$$V(\theta) = V_{tdc}\left(1 + \frac{CR-1}{2}\{(1-\cos\theta) + \frac{1}{\varepsilon}\left(1 - \sqrt{1-\varepsilon^2(\sin\theta)^2}\right)\}\right) \quad (5)$$

Systems of equations [3] used to calculate pressure, temperature, work, heat transfer, heat leakage are given below

$$\frac{dP}{d\theta} = \frac{A+B+C}{D+E} \quad (6)$$

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$$\frac{dT_b}{d\theta}=\frac{\upsilon_b}{C_{pb}}\frac{\partial \ln\upsilon_b}{\partial \ln T_b}\frac{A+B+C}{D+E}-h\frac{\left(\frac{\pi b^2}{2}+4\frac{V}{b}\right)}{\omega m C_{pb}}(T_b-T_w)\frac{\sqrt{x}}{x}+\frac{h_u-h_b}{xC_{pb}}\left(\frac{dx}{d\theta}+(x^2-x)\frac{C}{\omega}\right) \quad (7)$$

$$\frac{dT_u}{d\theta}=\frac{\upsilon_u}{C_{pu}}\frac{\partial \ln\upsilon_u}{\partial \ln T_u}\frac{A+B+C}{D+E}-h\frac{\left(\frac{\pi b^2}{2}+4\frac{V}{b}\right)}{\omega m C_{pu}}(T_u-T_w)\frac{1-\sqrt{x}}{1-x} \quad (8)$$

$$\frac{dW}{d\theta}=P\frac{dV}{d\theta} \quad (9)$$

$$\frac{dQ_l}{d\theta}=\frac{h}{\omega}\left(\frac{\pi b^2}{2}+4\frac{V}{b}\right)[\sqrt{x}(T_b-T_u)+(1-\sqrt{x})(T_u-T_w)] \quad (10)$$

$$\frac{dH_l}{d\theta}=\frac{Cm}{\omega}[h_u(1-x^2)-h_b x^2] \quad (11)$$

The coefficients A, B, C, D and E are used to simplify the equation layout [4].

2.2 Heat Transfer Models

Heat transfer coefficient h was calculated using 5 separate models.

2.2.1 Anand

Anand developed the following convection heat transfer correlation [3]

$$N_u=\frac{h_c B}{k}=a\left(\frac{\rho S_p B}{\mu}\right)^b \quad (12)$$

The value of 'a' is varied with intensity of charge motion and engine design. With normal combustion, $0.35<a<0.8$, $b=0.7$. 'a' is increased with increasing intensity of charge motion. Here $a=0.5$ was used.

2.2.2 Woschni

A well known correlation widely used to estimate instantaneous heat transfer coefficient is given by Woschni [4]

$$h=3.26T^{-.53}B^{-.2}P^{.8}w^{.8} \quad (13)$$

Where,

$$w=\left[C_1 S_p+C_2\frac{V_d T_r(P-P_m)}{P_r V_r}\right] \quad (14)$$

For compression: $C_1=2.28, C_2=0$

For combustion and expansion: $C_1=2.28, C_2=3.24\times10^{-3}$

Here 'w' is the local average cylinder gas velocity. During combustion and expansion, as gas velocity raises due to change in density, a term proportional to the pressure rise due to combustion ($P-P_m$) was added (P_m= motored cylinder pressure).Radiation effects are included in a lumped form during combustion and expansion and swirling effect is neglected.

2.2.3 Modified Woschni

$$h=\alpha_{scalling}L(t)^{-.2}P(t)^{.8}T^{-.73}\left[C_1 V_p+\frac{C_2 V_s T_r}{6P_r V_r}(p-P_m)\right]^{.8} \quad (15)$$

Characteristic length, $L(t)=\frac{4V}{\pi d_p^2}$

which means L(t) is the instantaneous chamber height

Here $\alpha_{scalling}$ is used to suit the particular engine geometry [5] and it's value is assumed 1000 here.

2.2.4 Hohenberg

The next correlation was developed by Hohenberg[6] for heat transfer coefficient.

$$h=C_1 V^{-.06}P^{0.8}T^{-0.4}\left(V_p+C_2\right)^{.8} \quad (16)$$

where, $V=\pi/6\ d_s^3$;(d_s-characteristic length)

$C_1=130$, $C_2=1.4$

Hohenberg model gives better prediction of heat fluxes where characteristic length is taken based on instantaneous cylinder volume instead of bore

2.2.5 Han et al

Han et al [7] developed the following empirical formula to determine heat transfer coefficient in SI engine.

$$h=C_1 p^{.75}U^{.75}B^{-.25}T^{-.465} \quad (17)$$

$$U=C_2 V_p+C_3\left(k\frac{PdV}{d\theta}+V\frac{dP}{d\theta}\right) \quad (18)$$

$C_1=687$, $C_2=0.494$, $C_3=7.3\times10^{-7}$, $k=1.35$

Here term 'U' carries the significance of gas velocity variations due to combustion which increase due to heat released from chemical reaction of gas mixture.

2.3 Simulation

Simulation was done based on [1] using a step size of 1°.Computer code was written in MATLAB.

2.3.1 Considerations

There are some considerations made before simulation. These are

- Considered the burned and unburned zones in the cylinder separately.
- Wall temperature was considered constant.
- Didn't simulate the gas exchanging process.
- valve geometry and timing wasn't taken into considerations.

- Burn duration was kept fixed
- ODE23 solver was used in MATLAB to solve DEQ.
- Residual mass fraction was kept constant.

2.3.2 Input parameters

Table.1: Input parameters.

parameters	**values**
Bore, B	3 inch
Start of burning, θ_s	35/8 inch
Burn Duration, θ_b	-30°
Half stroke to rod ratio, ε	0.25
Wall surface temperature, T_{wall}	420 K
Inlet Fuel temperature, T_1	310 K
Atmospheric pressure, P_1	101.325KPa
Blow by constant, $C_{blow\ by}$	1.1

3. Results and Discussions

3.1 Comparison with Experimental Results

Here various heat transfer models were analyzed to obtain the instantaneous spatial-averaged heat transfer coefficients and the effect of each model on various parameter change like pressure, temperature, heat transfer, work, heat flux etc. with respect to crank angle were observed.Only experimental values for pressure were available to compare with simulation results. Pressure obtained by these models was then compared with experimental values. If the pressures matches it can be assumed that other parameters also satisfies actual values.In the comparative pressure graphs of all the models the experimental data were taken from the measurements done by 'Rakopoulos'. [8]

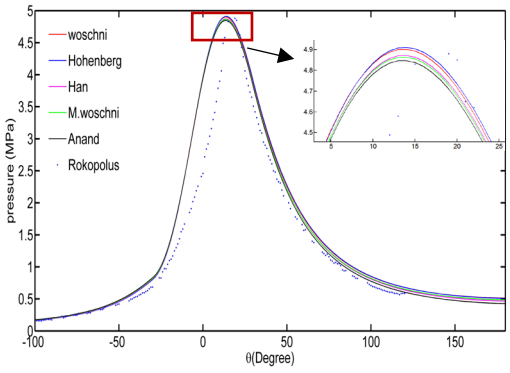


Fig.1: variation of pressure with crank angle

- All the models deviate from Rakopoulos [7] during compression. And it's an obvious situation because Rakopoulos [7] data doesn't provide precise results during compression stroke rather it underestimates the actual value.
- During compression, Hohenberg is the closest to experimental data. The sequence of the rest of the models are Woschni, Hanetal, Modified Woschni, Anand.
- During combustion, there is a phase lag of experimental value from obtained values. Here the peak value is slightly above Han et al and below Woschni. Anand under predicts and Hohenberg over predicts the peak value.
- Anand and Modified Woschni models are close to experimental during expansion.

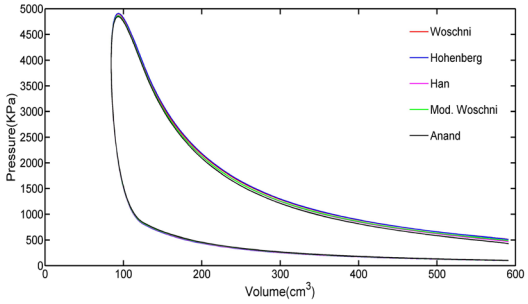


Fig.2: variation of pressure with cylinder volume

Temperature of burned and unburned zones are plotted separately in the fig 3.

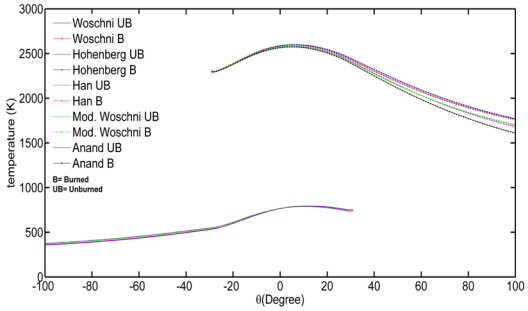


Fig.3: variation of temperature with crank angle

Work increases negatively during the compression stroke due to piston work on gas and then starts rising up substantially after the start of ignition. All the models are pretty closer to each other

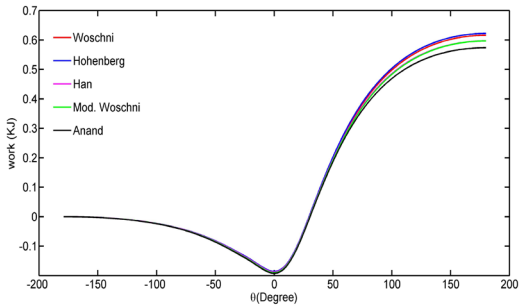


Fig.4: variation of work with crank angle

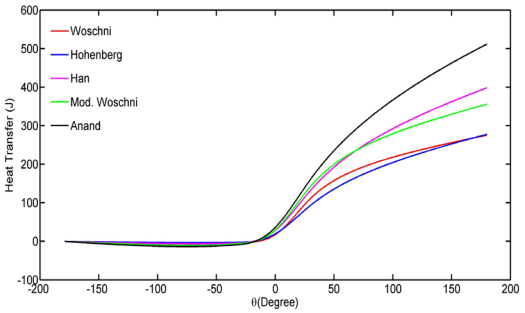


Fig.5: variation of heat transfer with crank angle

Variation of Heat flux can be summarized as follow

- Anand gives higher Heat Flux than any other model during combustion and expansion but it gives lower value during compression.
- Woschni under predicts in compression and expansion but overestimates the maximum value of the heat flux induced by combustion. Hohenberg can avoid this problem.

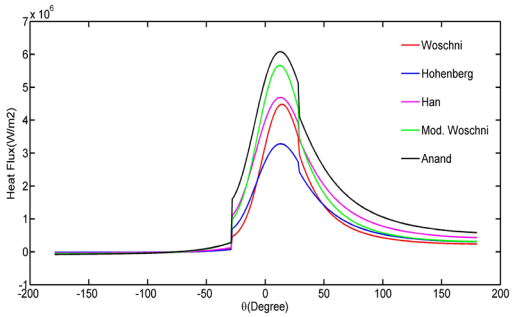


Fig 6: variation of heat flux with crank angle

- Han *et al* overestimates the heat flux during compression, underestimates during expansion and gives better prediction during combustion.
- Modified woschni gives better result in expansion. It is only applicable during compression and expansion since the spatially averaged heat flux model relies on the assumption of a closed control volume.

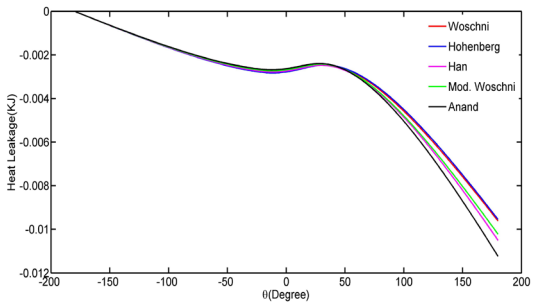


Fig.7: variation of heat leakage with crank angle

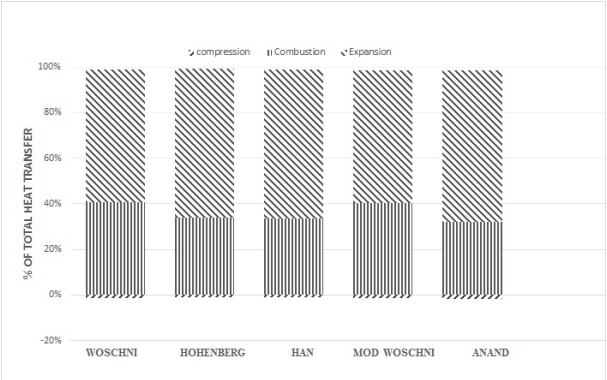


Fig.8: Bar graph showing amount of heat transfer in different stages for all models.

3.2 Results for different set of parameters

Simulations were made for some other sets of parameters to verify that the methodology is applicable for different parameters too.

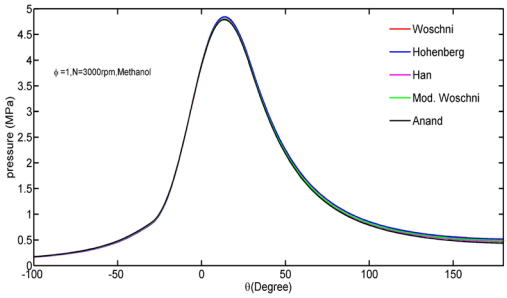


Fig.9: P-θ curve for phi=1,N=3000rpm, Fuel= methanol.

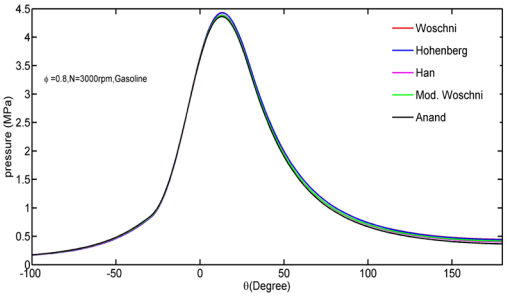


Fig.10: P-θ curve for phi=0.8,N=3000rpm, Fuel= Gasoline.

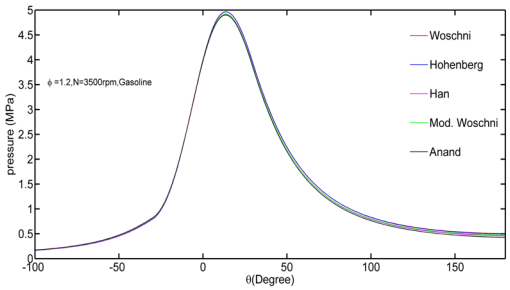


Fig.11: P-θ curve for phi=1.2,N=3500rpm, Fuel= Gasoline.

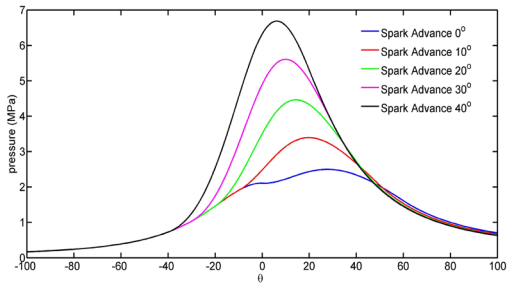


Fig.12: variation of pressure curve with the change in spark avance; model= woschni

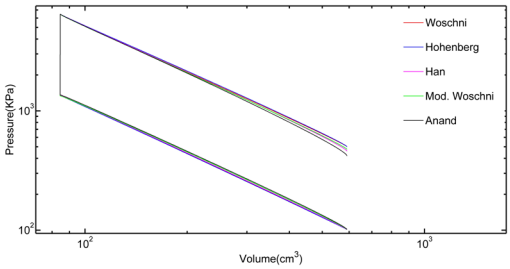


Fig.13: log-log plot of pressure variation with cylinder volume for spark at TDC , Fuel=Gasoline

4. Conclusion

Simulations were made to predict heat transfer of SI engine using different models discussed before and were compared with available experimental results. Besides percentage of heat transfer occurring in different stages for different models were shown in bar graph of fig 8. The bar chart shows Anand gives highest percentage of heat transfer in both compression and expansion whereas modified Woschni gives highest percentage in combustion. Model recommendations can be summarized as below from the results:

- Hohenberg model for compression stage.
- Han and Woschni model for combustion stage.
- Modified Woschni and Anand model for expansion stage.

There are scopes of improvements too. In all the models we assumed that wall temperature remains constant. But it can be varied to get better results. Besides radiation heat transfer should be considered properly to predict better results, which may require to include (T_G^4-T_w^4) term in the model.

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NOMENCLATURE

BDC	:	Bottom dead center
TDC	:	Top dead center
CR	:	Compression ratio
C	:	Coefficient for mass lost by blow-by
C_p	:	Specific Heat at constant pressure, $J.Kg^{-1}.K^{-1}$
H	:	Enthalpy, $J.Kg^{-1}$
h	:	Heat transfer coefficient, $W.m^{-2}.K^{-1}$
L	:	Rod length, m
M	:	Mass, Kg
P	:	Pressure, KPa
P_m	:	Motoring pressure, KPa
Q	:	Energy transferred as heat, J
S	:	Stroke length, m
T	:	Temperature, K
V	:	Instantaneous volume of combustion chamber, m^3
V_p	:	Piston speed, $m.s^{-1}$
x	:	Mass fraction burned
υ	:	Specific volume, $m^3.Kg^{-1}$
ε	:	Half stroke to rod ratio
θ	:	Crank angle, degree
ω	:	Rate of angular rotation, $rad.s^{-1}$
φ	:	Equivalence ratio
H_l	:	Heat leakage, J
W	:	Work, J
T_b	:	Temperature of burned zone, K
T_u	:	Temperature of unburned zone, K
T_w	:	Wall surface temperature, K
T_G	:	Gas temperature, K
Q_l	:	Heat transfer, J
Nu	:	Nusselt number
K	:	Thermal conductivity, $W.m^{-1}.K^{-1}$
σ	:	Stefan –boltzmann constant , $W.m^{-2}.K^{-1}$

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